Conflict Intensity and Global Search Attention

Evidence from the Russia–Ukraine War

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Abstract

We examine whether weekly changes in Russia–Ukraine conflict intensity predict worldwide Google search attention for *"ukraine war"* over 2023–2026 ($n = 171$ weekly observations), using an Autoregressive Distributed Lag model with Newey–West standard errors. We find a statistically significant one-week transmission from conflict to attention (HAC $p = 0.015$, cumulative IRF $= 0.388$ SD), driven entirely by attacks on Russian territory: a one-standard-deviation Russia-territory shock raises attention by 0.487 SD, nearly three times the Ukraine-territory response (0.177 SD). This asymmetry holds across politically opposed audiences; pro-Russia populations respond more strongly to Russia-territory attacks than pro-Ukraine populations do, ruling out confirmation bias. We interpret the pattern as information normalisation: after three years of continuous warfare, combat in Ukraine has become the expected baseline, while attacks on Russian soil remain expectation-violating. The same mechanism extends to financial markets: Russia-territory attacks Granger-cause changes in the VIX ($p = 0.016$), while Ukraine-territory attacks do not ($p = 0.698$).

1. Introduction

When does a war stop being news? Three years into the Russia–Ukraine conflict, the answer appears to be: when it happens in the expected place. This paper investigates whether weekly fluctuations in conflict intensity, measured by the number of violent events recorded by the Armed Conflict Location and Event Data (ACLED), explain fluctuations in worldwide Google search interest for the term *"ukraine war."* Using 171 weekly observations from January 2023 to March 2026, we find that they do, but with a striking asymmetry: it is attacks on *Russian* territory, not Ukrainian territory, that drive global attention.

This finding reflects a central question in the economics of information: under rational inattention (Sims, 2003), people allocate limited attention by processing some signals selectively rather than absorbing all available news. Applied to conflict, the implication is that audiences may habituate to repeated combat in Ukraine, while remaining sensitive to rarer and expectation-violating attacks on Russian territory.

We make three contributions. First, we document a statistically significant dynamic relationship between conflict intensity and global search attention using an autoregressive distributed lag model with Newey–West standard errors and bootstrap impulse-response bands. Second, we show that this relationship is driven by the *location* of conflict, not simply its magnitude, and holds across politically opposed audience groups. Third, we extend the analysis to financial markets and show that Russia-territory attacks also predict changes in the VIX, suggesting that the same normalisation mechanism appears in market uncertainty.

2. Literature Review

Our analysis builds on three strands of literature. First, the rational inattention framework (Sims, 2003) predicts that agents allocate attention to signals with high informational content relative to prior beliefs. Second, Google Trends has been widely used as a proxy for public and investor attention by Da et al. (2011), who study attention in equity markets; our setting differs in that we examine habituation during a prolonged ongoing event rather than information discovery. Third, studies of geopolitical risk show that conflict-related news can move financial markets (Rigobon & Sack, 2005; Caldara & Iacoviello, 2022). Our contribution is to show that within one ongoing war, the *location* of conflict events matters more than their frequency for both public attention and market uncertainty.

3. Data

3.1. Conflict intensity: ACLED

We measure conflict intensity using data from the Armed Conflict Location and Event Data project (Raleigh et al., 2010), restricting the sample to three event types: Battles, Explosions/Remote violence, and Violence against civilians. Events are aggregated to weekly frequency and split by country of occurrence (Ukraine or Russia), yielding `ukraine_events`, `russia_events`, and a combined total. The sample runs from January 2023 to March 2026 ($n = 171$ weeks); 2022 is excluded due to a normalisation discontinuity in the stitched Google Trends series (Appendix Figure A1).

3.2. Public attention: Google Trends

We measure public attention using the GT weekly index scaled 0–100 for *"ukraine war"* at the worldwide and country level. To limit scaling inconsistencies, we download data year-by-year and stitch the series. GT labels each weekly observation by the Sunday starting date of the week, whereas ACLED uses the Saturday ending date; we therefore shift GT dates forward by six days before merging.

For the audience heterogeneity extension, we construct two country groups based on citizen-level public opinion surveys rather than government positions, since our outcome variable is public search behaviour. The **pro-Ukraine group** comprises Poland, Sweden, the United Kingdom, and Estonia; the **pro-Russia group** comprises Serbia, India, Vietnam, and Indonesia (Ipsos, 2025; Pew Research Center, 2023; YouGov, 2024; Friedrich-Ebert-Stiftung, 2025). Both group series are stationary in first differences (Appendix Table A1).

3.3. Financial variables and transformations

Weekly VIX data are obtained from FRED and weekly S&P 500 closing prices from Yahoo Finance, both downloaded programmatically. All conflict and attention series are first-differenced to remove non-stationarity (confirmed by ADF and KPSS tests; Appendix Table A1) and then z-score standardised, placing both on a common scale so that coefficients and impulse responses are interpretable in standard deviation units. For the financial extension, we use first differences of the VIX and weekly S&P 500 returns (Appendix Figure A2). Seasonality checks confirm no significant monthly pattern in either series (ΔGoogle Trends $p = 0.462$; Δconflict $p = 0.688$; Appendix Figure A3).

4. Methodology

4.1. ADL model

Our primary specification is an Autoregressive Distributed Lag (ADL) model:

$$\Delta g_t \;=\; \alpha + \phi_1 \Delta g_{t-1} + \phi_2 \Delta g_{t-2} + \sum_{k=0}^{2} \beta_k \Delta c_{t-k} + \varepsilon_t \tag{1}$$

where Δg_t is the standardised weekly change in Google Trends and Δc_t is the standardised weekly change in conflict events. The model is denoted **m4b**. Own lags capture persistence in search attention, while conflict lags capture the dynamic transmission from conflict to attention.

Model selection compares five specifications by AIC and BIC (Appendix Table A2). AIC selects m4 (two conflict lags); BIC selects the own-lag-only benchmark. We prefer AIC at $n \approx 167$ with a weak signal. We augment m4 with a second own lag after the Breusch–Godfrey test rejects no-autocorrelation in m4 residuals ($p = 0.046$); the resulting m4b specification passes the same test ($p = 0.121$). Standard errors are corrected using Newey–West HAC (bandwidth 4). ACF and PACF diagnostics for both series are reported in Appendix Figure A4; residual diagnostics for m4b are shown in Appendix Figure A5. As a robustness check, we also estimate the same dynamic specification as an ARMA-X(2,0,2) model by Maximum Likelihood.

4.2. VAR and Granger causality

For the financial markets extension, we estimate a trivariate VAR(1) in $(\Delta c_t, r_t^{\text{SP500}}, \Delta \text{VIX}_t)$, separately for Ukraine- and Russia-territory conflict. We apply a Cholesky decomposition with ordering conflict → S&P 500 → VIX: conflict events are determined outside markets; equity returns respond contemporaneously; the VIX adjusts to both. We acknowledge that the VIX is forward-looking and could in principle react simultaneously with equity prices; placing it last is therefore a conservative choice. If anything, this ordering makes it harder to attribute VIX movements to conflict, so any significance we find constitutes a lower bound on the true effect. Bootstrap confidence bands (95%, 500 replications) are computed using `vars::irf()`.

5. Results and Discussion

5.1. Conflict drives global search attention

Table 1 reports estimates for the preferred model m4b. The key finding is the coefficient on Δc_{t-1}: a one-standard-deviation increase in conflict intensity raises worldwide search attention by 0.183 SD one week later ($p = 0.015$, HAC). This result survives correction for heteroskedasticity and autocorrelation. After re-estimating the same dynamic specification by Maximum Likelihood ARMA-X(2,0,2), we find the cumulative impulse response virtually unchanged: 0.388 SD under the OLS-ADL model versus 0.385 SD under ARMA-X (Appendix Table A3). We therefore keep the OLS-ADL specification as the baseline because it is more transparent while delivering the same dynamic conclusion. Figure 1 reports the corresponding impulse response. No reverse Granger causality is found: search attention does not predict conflict ($p = 0.151$).

Table 1. Preferred model m4b — OLS and HAC estimates

	OLS		**HAC (NW, BW = 4)**	
Variable	Coef.	SE	Coef.	SE
Constant	0.005	(0.077)	0.005	(0.067)
Δg_{t-1}	−0.010	(0.078)	−0.010	(0.116)
Δg_{t-2}	−0.132	(0.078)*	−0.132	(0.084)
Δc_t	0.122	(0.083)	0.122	(0.107)
Δc_{t-1}	**0.183**	(0.089)**	**0.183**	(0.078)**
Δc_{t-2}	0.138	(0.085)	0.138	(0.090)
$N = 166$	$R^2 = 0.049$	AIC = 477.81	BIC = 499.59	BG $p = 0.121$

*$p < 0.10$, **$p < 0.05$. All series first-differenced and z-score standardised. NW = Newey–West; BG = Breusch–Godfrey.

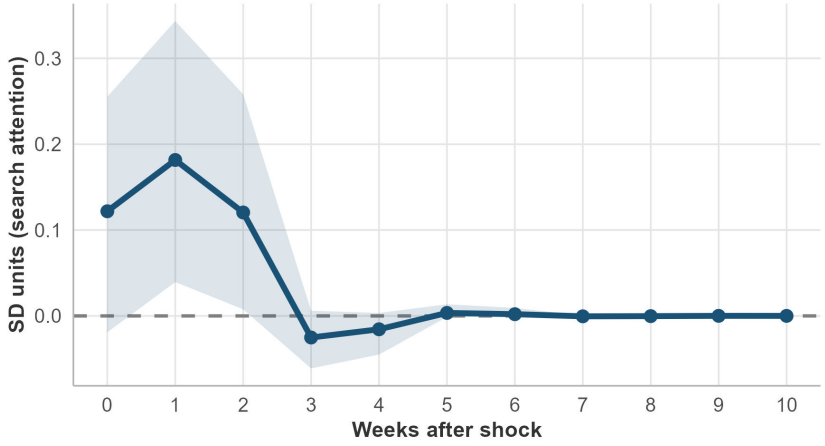


Figure 1. IRF: response of worldwide Google search attention to a one-SD shock in combined conflict intensity (m4b). Shaded band = 90% bootstrap CI (500 replications). Response peaks at week 1 and dissipates by week 4. Cumulative effect: 0.388 SD.

5.2. The normalisation hypothesis: location dominates magnitude

We split the conflict series by territory of occurrence. Figure 2 overlays the IRFs for Ukraine- and Russia-territory shocks. The cumulative response to a one-SD Russia-territory shock is **0.487 SD**, compared to **0.177 SD** for a Ukraine-territory shock. AIC and BIC both select the Russia-only model over the Ukraine-only specification (AIC: 475.98 vs 484.50; BIC: 497.81 vs 506.33), further supporting the normalisation hypothesis.

We interpret this asymmetry as evidence of information normalisation, or desensitisation to repeated conflict news. After more than three years of continuous, high-intensity warfare in Ukraine, global audiences have formed strong priors about combat occurring on Ukrainian territory: it is expected and therefore generates little marginal search activity. Attacks on Russian territory, by contrast, are rarer and more informationally salient, driving disproportionate search behaviour. The informational content of an event, not its absolute severity, determines the attentional response.



Figure 2. IRF by conflict territory: Ukraine-territory shock (blue, solid) vs Russia-territory shock (red, dashed) on worldwide Google search attention. Shaded bands = 90% bootstrap CI. Cumulative effects: Ukraine = 0.177 SD; Russia = 0.487 SD.

5.3. Audience heterogeneity: surprise dominates alignment

A natural concern is whether the Russia-territory effect reflects confirmation bias rather than normalisation. Under this alternative, we would expect pro-Russia audiences to respond more to Russia-territory attacks and pro-Ukraine audiences to respond more to Ukraine-territory attacks. Our results reject this cleanly.

Table 2 reports HAC p-values and R^2 for all six audience × territory models. Ukraine-territory attacks are statistically insignificant in every audience group. Russia-territory attacks are significant in both the pro-Russia group ($p = 0.015$, Model F) and the pro-Ukraine group ($p = 0.047$, Model D), and marginally significant worldwide ($p = 0.051$, Model B). Crucially, the strongest lag-1 response is found in Model F, the opposite of what a simple confirmation-bias story would predict. This finding is noteworthy: if anything, pro-Russia audiences, who might be expected to discount attacks on Russian territory as propaganda, respond *more* strongly than pro-Ukraine audiences, suggesting that the attention effect is driven by genuine information surprise rather than ideological reinforcement. Appendix Figure A7 plots the full IRF paths for all six models, confirming that the Russia-territory response dominates consistently across every audience group.

Table 2. Audience heterogeneity models — fit and HAC inference ($n = 170$)

Model	Audience	Conflict location	Lag-1 (SD)	R^2	HAC p
A	Worldwide	Ukraine	0.075	0.030	0.179
B	Worldwide	Russia	0.229	0.077	$0.051^{\dagger}$
C	Pro-Ukraine	Ukraine	0.084	0.049	0.229
D	Pro-Ukraine	Russia	**0.225**	0.069	$\mathbf{0.047}^{**}$
E	Pro-Russia	Ukraine	0.060	0.037	0.314
F	Pro-Russia	Russia	**0.265**	**0.090**	$\mathbf{0.015}^{**}$

$^{\dagger}p < 0.10$, $^{**}p < 0.05$. Lag-1 HAC coefficient in SD units. Pro-Ukraine: Poland, Sweden, UK, Estonia. Pro-Russia: Serbia, India, Vietnam, Indonesia (Ipsos, 2025; Pew Research Center, 2023; YouGov, 2024; Friedrich-Ebert-Stiftung, 2025).

5.4. Financial markets: normalisation in the VIX

We extend the analysis to financial markets using a trivariate VAR with S&P 500 returns as a control. Table 3 reports the key Granger-causality results. Russia-territory attacks Granger-cause changes in the VIX after controlling for equity returns ($p = 0.016$), while Ukraine-territory attacks do not ($p = 0.698$). No reverse causality is found: VIX changes do not predict conflict ($p = 0.997$).

These results suggest that markets, like the global public, respond mainly to expectation-violating attacks on Russian territory rather than continued combat in Ukraine. This is consistent with Caldara & Iacoviello (2022), who show that geopolitical risk spikes are driven by unexpected escalation. The corresponding VIX impulse responses are reported in Appendix Figure A6.

Table 3. Financial-market extension: Granger-causality evidence

Test	p-value	Conclusion
Ukraine-territory conflict → VIX	0.698	Not significant
Russia-territory conflict → VIX	**0.016**	**Significant**
VIX → conflict	0.997	Not significant

6. Discussion

Taken together, our results paint a coherent picture: information surprise, not conflict magnitude, determines both public attention and financial uncertainty during prolonged warfare. A one-standard-deviation Russia-territory shock raises worldwide search attention by 0.487 SD and significantly moves the VIX; Ukraine-territory effects are statistically weak in the search models and insignificant in the VIX extension. The conflict has entered a phase where combat in Ukraine has been fully absorbed by global priors, while attacks on Russian soil remain expectation-violating, precisely what rational inattention predicts (Sims, 2003).

Two limitations merit acknowledgement. First, Google Trends indices are re-normalised at each download, introducing possible scaling inconsistencies across yearly files; Newey–West standard errors only partially address this. Second, ACLED may underreport events on Russian territory due to asymmetric media access, meaning our Russia-territory estimates are conservative lower bounds.

Conclusion. Conflict normalisation is detectable in both public search behaviour and financial markets: three years in, it is not the intensity of fighting but its location that moves global attention and market uncertainty. Future work could sharpen these estimates using higher-frequency data or an event-study design centred on specific attacks on Russian territory.

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Appendix A — Data and Diagnostics

Table A1. Stationarity tests — levels and first differences

	Levels		**First differences**	
Series	ADF p	KPSS p	ADF p	KPSS p
Main analysis				
`combined_events`	0.540	0.010	0.010	0.100
`ukraine.war`	0.033	0.016	0.010	0.100
Split conflict series				
`ukraine_events`	—	—	0.010	0.100
`russia_events`	—	—	0.010	0.100
Audience heterogeneity				
`gt_ukraine_group`	—	—	0.010	0.100
`gt_russia_group`	—	—	0.010	0.100
Financial extension				
`vix`	0.037	0.038	0.010	0.100
`r_sp500`	—	—	0.010	0.100

ADF null: unit root. KPSS null: stationary. All series are treated as stationary in first differences. "—" indicates that levels tests were not formally reported for series used only in first-differenced form.

Table A2. ADL model selection — AIC and BIC ($n = 167$)

Model	Specification	k	AIC	BIC
m1	Own lag only	2	481.50	**490.85**
m2	Own lag + conflict lag 0	3	482.87	495.35
m3	Own lag + conflict lags 0–1	4	481.55	497.14
m4	Own lag + conflict lags 0–2	5	**480.95**	499.65
m5	Own lag + conflict lags 0–3	6	482.94	504.76

Table A3. ARMA-X robustness check — Maximum Likelihood estimates

Parameter	Estimate	Std. error	t-ratio
Constant	0.005	0.076	0.064
ϕ_1	-0.010	0.077	-0.134
ϕ_2	-0.132	0.077	-1.717
σ	0.975	0.053	18.275
β_0	0.122	0.081	1.498
β_1	0.183	0.087	2.096
β_2	0.138	0.083	1.660

ARMA-X(2,0,2) estimated by Maximum Likelihood. The cumulative IRF is 0.385 SD, compared to 0.388 SD under the baseline OLS-ADL specification.

Appendix B — Supplementary Figures

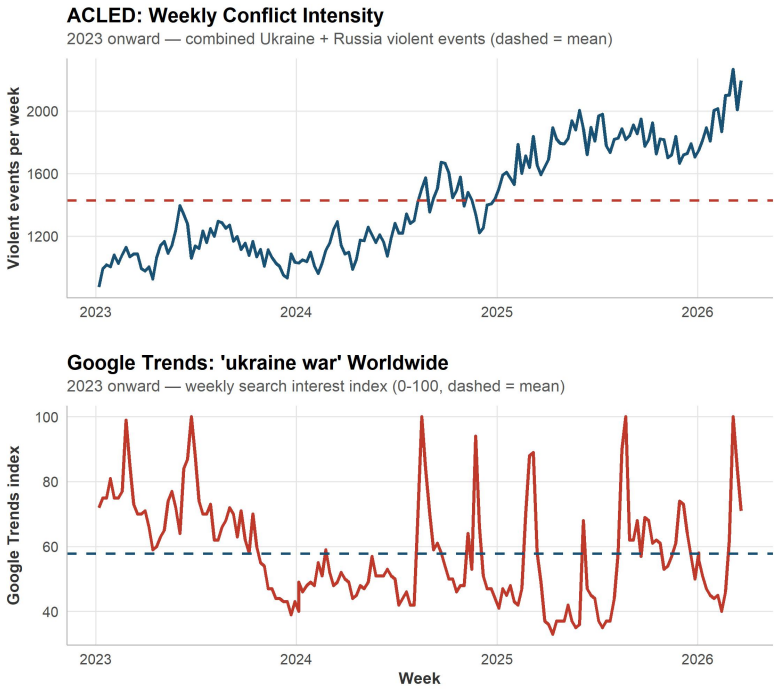


Figure A1. Raw weekly series: combined ACLED conflict intensity in Ukraine and Russia (top) and worldwide Google Trends search interest for "ukraine war" (bottom). Dashed lines indicate sample means.

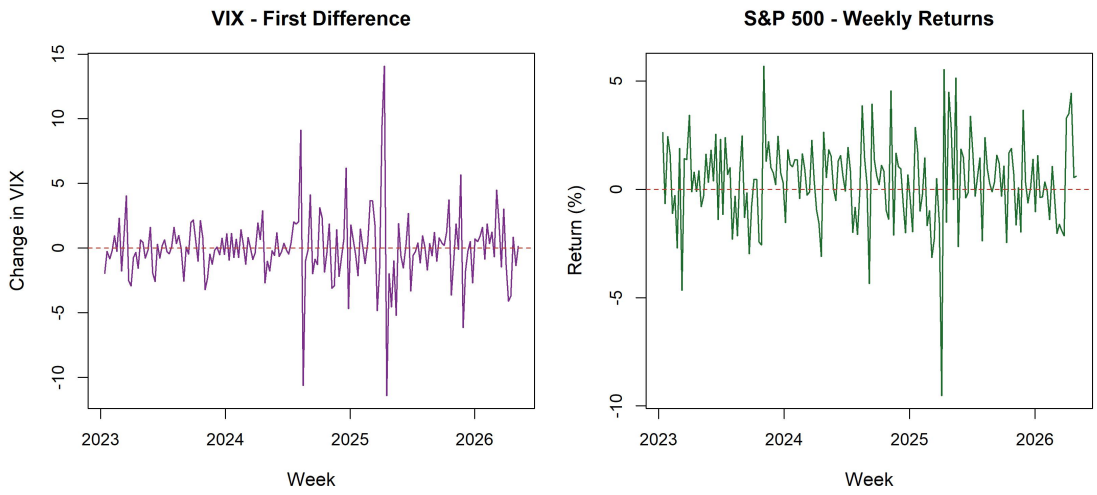


Figure A2. Financial variables used in the VAR extension: first differences of the VIX and weekly S&P 500 returns. Dashed lines indicate zero.

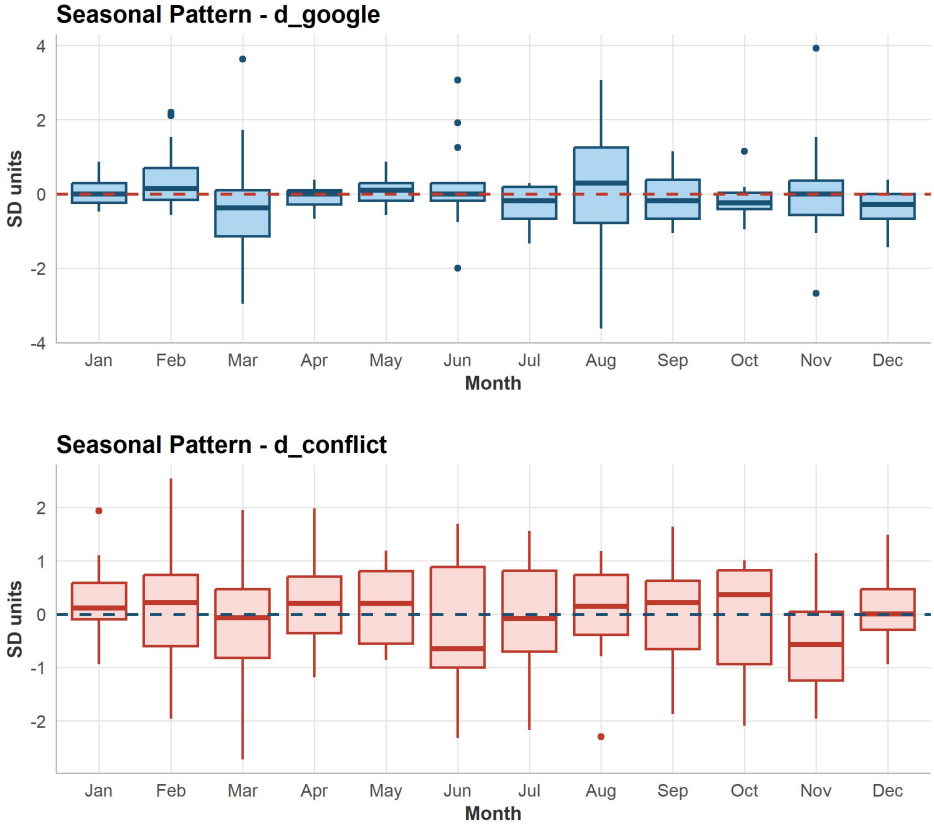
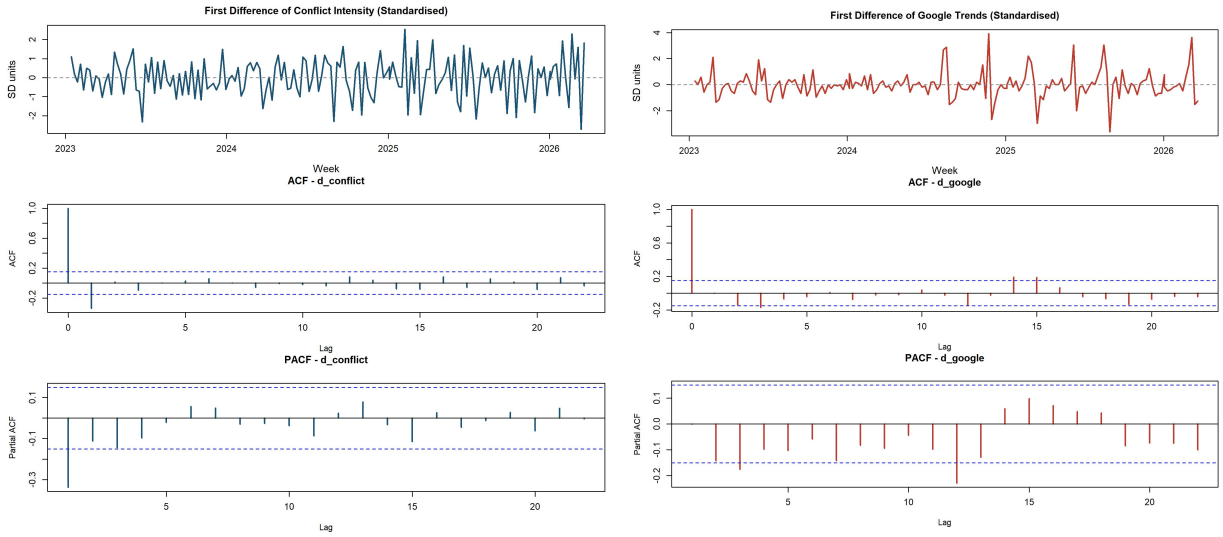


Figure A3. Seasonality check for the first-differenced, standardised Google Trends and conflict series. Monthly boxplots show no clear seasonal pattern; F-tests for monthly effects are insignificant for both ΔGoogle Trends ($p = 0.462$) and Δconflict ($p = 0.688$).



(a) Δ conflict

(b) Δ Google Trends

Figure A4. ACF and PACF diagnostics for the first-differenced, standardised conflict and Google Trends series. Δconflict shows a strong negative spike at lag 1, while ΔGoogle Trends shows no dominant autocorrelation structure.

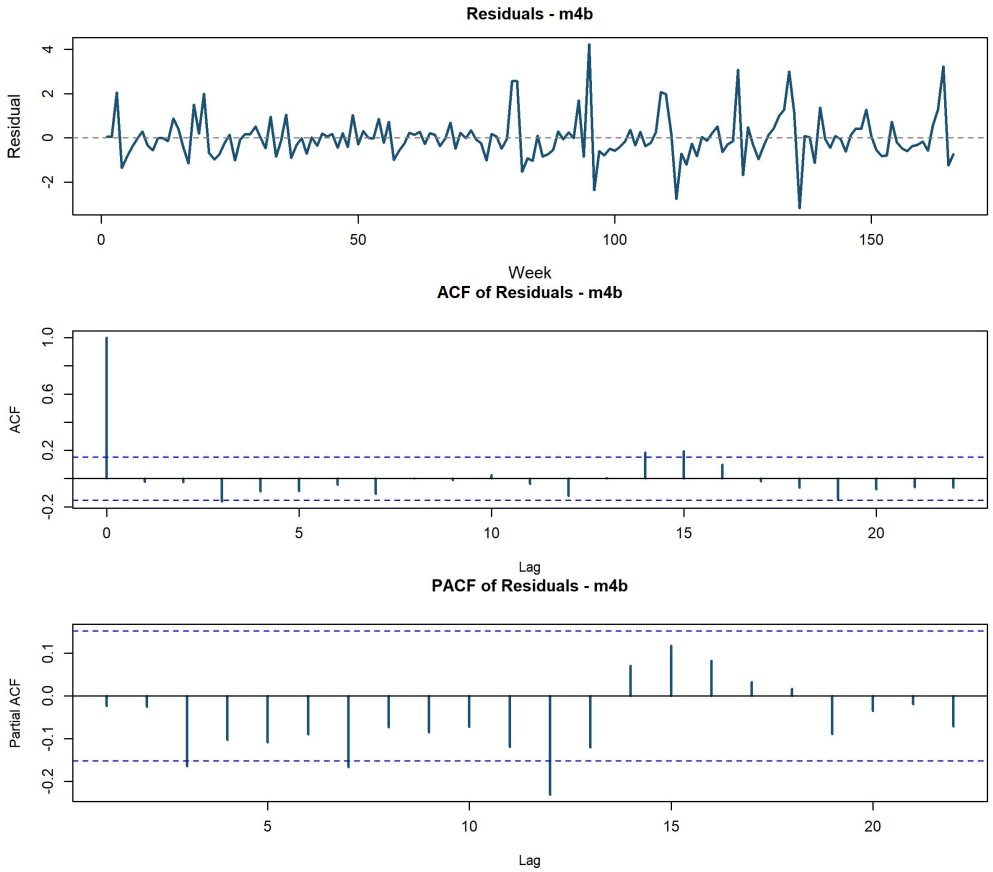


Figure A5. Residual diagnostics for m4b. Breusch–Godfrey $p = 0.121$.

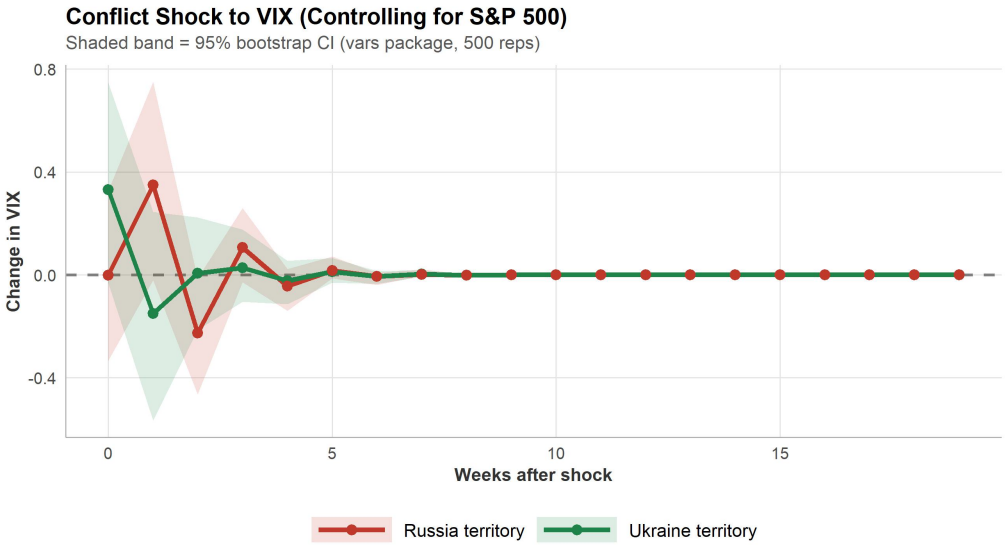


Figure A6. VIX impulse responses to Ukraine-territory and Russia-territory conflict shocks, controlling for S&P 500 returns. Shaded bands = 95% bootstrap CI (`vars::irf()`, 500 replications).



Figure A7. IRFs for all six audience × conflict-location models with 90% bootstrap bands. Russia-territory models consistently dominate Ukraine-territory models across all audience groups.